Lab 7

Problema 1

$S \to 0A0 \mid 1B1 \mid BB$

$A \to C$

$B \to S \mid A$

$C \to S \mid \varepsilon$

1). Epsilon

$S \to 00 \mid 1B1 \mid BB \mid 0A0$ $\quad S \to 0A0 \mid 11 \mid 1B1 \mid B$

$= S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$

$A \to C$

$B \to S \mid A$

$C \to S$

2) unarias

$P_0 = \{(S,S),(A,A),(B,B),(C,C)\}$

(S,S)	$S \to B$	(S,B)
(S,B)	$B \to A$	(S,A)
(S,A)	$A \to C$	(S,C)
(S,C)	$C \to S$	(S,S)

(B,B)	$B \to A$	(B,A)
(B,A)	$A \to C$	(B,C)
(B,C)	$C \to S$	(B,S)
(A,A)	$A \to C$	(A,C)
(A,C)	$C \to C$	(A,S)

(C,C) $\quad C \to S$ $\quad (C,S)$

(S,S) $\quad S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

(S,A); (S,B); (S,C) No generan nuevas opciones

(B,S) $\quad S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

(B,B); (B,A); (B,C) No generan nuevas opciones.

(A,S) $\quad S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

(A,A); (A,B); (A,C) No generan nuevas opciones.

(C,S) $S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

(C,C); (C,B); (C,A) No generan nuevos opciones

$S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$A \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$B \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$C \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

3) Inutiles e Inalcanzables

$T = \{0,1\}$ $W_1 = \{S, A, B, C\}$ $W_2 = \{S, A, B, C\}$ No hay

$y_1 = \{S\}$ $y_2 = \{S, A, B, 0, 1\}$ $y_2 \{S, A, B, 0, 1\}$

$S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$A \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$B \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

4). Chomsky

$T_0 = 0$ $T_1 = 1$

$S \to T_0 A T_0 \mid T_1 B T_1 \mid 00 \mid 11 \mid BB$

$A \to T_0 A T_0 \mid T_1 B T_1 \mid 00 \mid 11 \mid BB$

$B \to T_0 A T_0 \mid T_1 B T_1 \mid 00 \mid 11 \mid BB$

$C_1 = A T_0$ $C_2 = B T_1$

$S \to T_0 C_1 \mid T_1 C_2 \mid 00 \mid 11 \mid BB$

$A \to T_0 C_1 \mid T_1 C_2 \mid 00 \mid 11 \mid BB$

$B \to T_0 C_1 \mid T_1 C_2 \mid 00 \mid 11 \mid BB$

Ejercicio 2:

S → aAa|bBb|ε
A → c|a
B → c|b
C → CDF|C
D → A|B|ab

1) Epsilon

S → aAa|aa|bBb
S → aAa|bb|bBb

S → aAa|aa|bb|bBb
A → c|a
B → c|b
C → CDE|DE|CF|E
D → A|B|ab

2) Unarias

(C,C) C → E (C,E)
(S,S)
(A,A) A → C (A,C)
(A,C) C → E (A,E)
(B,B) B → C (B,C)
(B,C) C → E (B,E)
(D,D) D → A (D,A)
(D,A) A → C (D,C)
(D,C) C → E (D,E)
(D,D) D → B (D,B)
(D,B)

(C,C): CDE|DE|CE
(S,S): aAa|aa|bb|bBb
(A,A): a
(A,C): CDE|DE|CE
(B,B): b
(B,C): CDE|DE|CE
(D,D): ab
(D,A): a

(C,E) = Nada

(D,B): b
(D,C): CDE|DE|CE
(D,E): Nada

S → aAa | aa | bb | bBb
A → CDE | DE | CE | a
B → CDE | DE | CE | b
C → CDE | DE | CE
D → CDE | DE | CE | ab | a | b

3). Inutiles

$T = \{a, b\}$ $W_1 = \{S, A, B, D, a, b\}$ $W_2 = \{S, A, B, C, D, a, b\}$
$W_3 = \{S, A, B, C, D, a, b\}$

$Y_1 \Rightarrow \{S\}$ $Y_2 = \{D, A, B, a, b, S\}$ $Y_3 \rightarrow \{D, A, B, a, b, C, S\}$

S → aAa | aa | bb | bBb
A → CDE | DE | CE | a
B → CDE | DE | CE | b
C → CDE | DE | CE
D → CDE | DE | CE | ab | a | b

4) chomsky

$C_1 = DE$ $T_1 = a$ $T_2 = b$
S ⇒ $T_1AT_1 \mid T_1T_1 \mid T_2T_2 \mid T_2BT_2$
A → $CC_1 \mid DE \mid CE \mid T_1$
B → $CC_1 \mid DE \mid CE \mid T_2$
C → $CC_1 \mid DE \mid CE$
D → $CC_1 \mid DE \mid CE \mid T_1T_2 \mid T_1 \mid T_2$

$C_2 = AT_1$ $C_3 = BT_2$
S → $T_1C_2 \mid T_1T_1 \mid T_2T_2 \mid T_2C_3$
A → $CC_1 \mid DE \mid CE \mid T_1$
B → $CC_1 \mid DE \mid CE \mid T_2$
C → $CC_1 \mid DE \mid CE$
D → $CC_1 \mid DE \mid CE \mid T_1T_2 \mid T_1 \mid T_2$

Ejercicio 3.

$S \to ASA \mid aB$

$A \to B \mid S$

$B \to b \mid \varepsilon$

1) Epsilon

$S \to ASA \mid a \mid aB$

$S \to SA \mid AS \mid ASA \mid aB$

$S \to ASA \mid SA \mid AS \mid aB \mid a \mid S$

$A \to B \mid S$

$B \to b$

2) unarias $P_0 = \{(S,S), (A,A), (B,B)\}$

$(A,A) \quad A \to B \quad (A,B)$

$(A,A) \quad A \to S \quad (A,S)$

(A,A)

$(A,S) \to ASA \mid AS \mid SA \mid aB \mid a$

$(A,B) \to b$

$S \to ASA \mid AS \mid SA \mid aB \mid a$

$A \to ASA \mid AS \mid SA \mid aB \mid a \mid b$

$B \to b$

3) Inutiles $T = \{a, b\}$ $W_1 = \{S, A, B, a, b\}$ $W_2 = \{S, A, B, a, b\}$

$Y_1 = \{S\}$ $Y_2 = \{S, A, a, b\}$ $Y_3 = \{S, A, B, a, b\}$ $Y_4 \{S, A, B, a, b\}$

$S \to ASA \mid AS \mid SA \mid aB \mid a$

$A \to ASA \mid AS \mid SA \mid aB \mid a \mid b$

$B \to b$

4 chomsky $C_1 = AS$ $T_1 = a$ $T_2 = b$

$S \to C_1A \mid AS \mid SA \mid T_1B \mid T_1$

$A \to C_1A \mid AS \mid SA \mid T_1B \mid T_1 \mid T_2$

$B \to T_2$